



Homework Review – Scores and ROC Chart

You are asked to evaluate the performance of two classifications models, M1 and M2. The test set you have chosen contains 26 binary attributes, labeled A through Z.

The table on the right shows the probabilities obtained by applying the models to the test set. Only the probabilities for the positive class are shown, as it is a two-class problem, the corresponding negative probabilities are $p(-, M1) = 1 - p(+, M1)$ and $p(-, M2) = 1 - p(+, M2)$.

Assume that we are mostly interested in detecting observations from the positive class.

Observation	True class	p(+, M1)	p(+, M2)
1	+	0.73	0.61
2	+	0.69	0.03
3	-	0.44	0.68
4	-	0.55	0.31
5	+	0.67	0.45
6	+	0.47	0.09
7	-	0.08	0.38
8	-	0.15	0.05
9	+	0.45	0.01
10	-	0.35	0.04

- Plot the ROC charts for M1 and M2 (on the same graph) and compute the AUCs. Which model do you think is better? Explain your answer!
- For model M1, assume you choose a threshold of $t=0.5$. In other words, any test instance whose probability is greater than t will be classified as positive. Compute the Accuracy, Precision and Recall, Sensitivity and Specificity and F_1 -Score.
- Repeat the same analysis for model M2 for the same threshold. Compare the F_1 -score of the two models. Which one is better? Is the result consistent with the ROC chart?
- Repeat the same analysis for model M1 with a threshold of $t=0.1$. Which threshold do you prefer, 0.5 or 0.1? Is the result consistent with what you would expect from the ROC chart?

Cheat Sheet:

$$\text{accuracy} = (TP+TN) / (TP+TN+FP+FN)$$

$$\text{precision} = TP / (TP+FP)$$

$$\text{TPR} = \text{recall} = \text{sensitivity} = TP / P = TP / (TP+FN)$$

$$\text{specificity} = TN / N = TN / (TN+FP)$$

$$\text{FPR} = FP / N = FP / (FP + TN)$$

$$F_1\text{-Score} = 2TP / (2TP+FP+FN)$$